

CAREER EPISODE 3

Automated night lamp

A. INTRODUCTION

CE 3.1

I took an initiative to propose an automated night lamp during my engineering academic project. For the design and development part I utilized a microcontroller (ATMEGA328P) for the interfacing of the LDR and relay module. Through my design methodology I efficiently designed an automated night lamp by which electricity could be saved through automated switching OFF the light at day time and switching ON the light at night time. I provided a buzzer in the system design for enabling the system to alarm on detecting the dawn and switch off automatically after a set time. I worked hard applying my knowledge and using the best techniques.

B. BACKGROUND

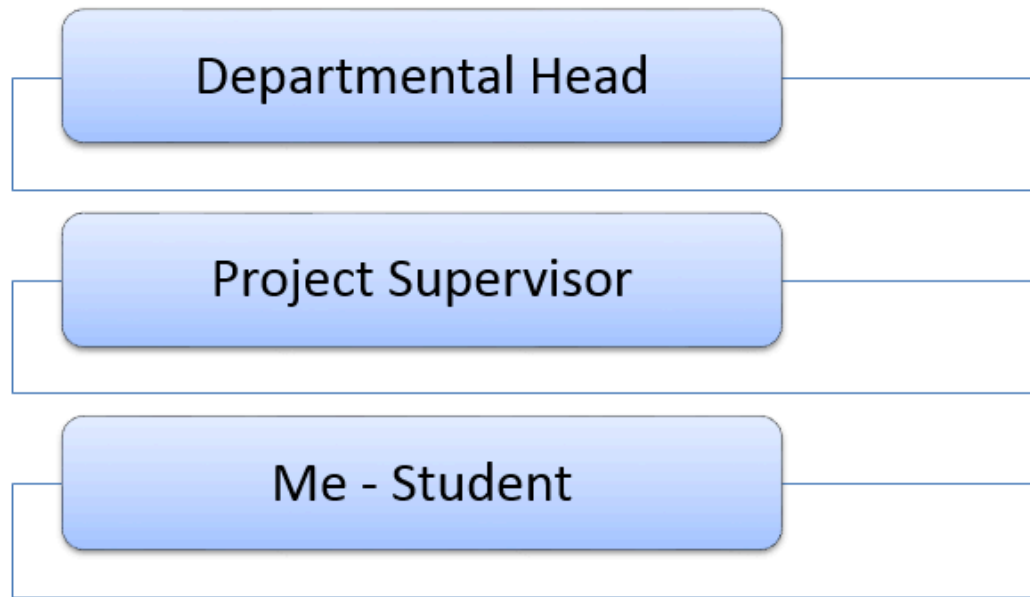
CE 3.2

For gaining knowledge and skills associated with my current project, I referred to research papers and relevant journals. I understood the application of the distinct methods and electronics engineering techniques through background study. With an aim of saving electric power I worked on designing an automated night lamp whereas to clearly understand the scope of current work, I thoroughly studied related systems and the framework. Accordingly I devised the methodology for proposing an automated lamp system with an integrated alarm wherein ATMEGA328P controller IC and LDR sensor were used. Using an electronic circuit design tool I completed the schematic design of the system by integrating different components and resources associated with electronics engineering. The methodology I framed included tasks for system design and the selection of components for the same. For circuit design, I chose the components systematically considering their relevance for the project and this helped accomplish the desired design and meet the goal, thereby project requirements.

CE 3.3

- ◆ Through this project work I addressed the issues related to the wattage of electric power and to save it I introduced an advanced automated night lamp using ATMEGA328P controller interfaced with LDR sensor.
- ◆ Using an electronic circuit design tool I completed the schematic design of the system by integrating different components such as LDR sensor, Relay, 555-timer IC, buzzer, resistors, LEDs, capacitors, transistors, diodes, ATMEGA328P controller IC etc. whereas the connected LED under the project acts as the night lamp.
- ◆ I also deployed a battery for backup and continuous supply to a power circuit.
- ◆ For the control of the system, an ATMEGA328P controller IC acts as the microcontroller which I implemented with proper pin-configuration and programming code that I developed in Arduino IDE using C language.
- ◆ I analyzed the circuit performance through the Proteus simulator and for this I applied the power needed to the designed circuit corresponding to which the connected circuit components operate.
- ◆ After this project prototype developed with assembly of hardware project components on breadboard

CE 3.4



C. PERSONAL ENGINEERING ACTIVITY

CE 3.5

For the purpose of saving electrical power I designed and implemented an advanced automated night lamp circuit which would turn-on or tune-off the lamp automatically on sensing the darkness in the room or a specific place. With this project concept I designed the system design in a manner that the lamp would remain in an ON state until the daylight would be sensed by the light sensor (i.e. LDR) in the morning, however to achieve the control mechanism I interfaced the LDR sensor with ATMEGA328P controller IC and also I utilized LED under the project as the night lamp which would provide bright light in the room. Technically, I specified the important components in the project design and prepared a block based functional diagram that I shown below in figure 1-

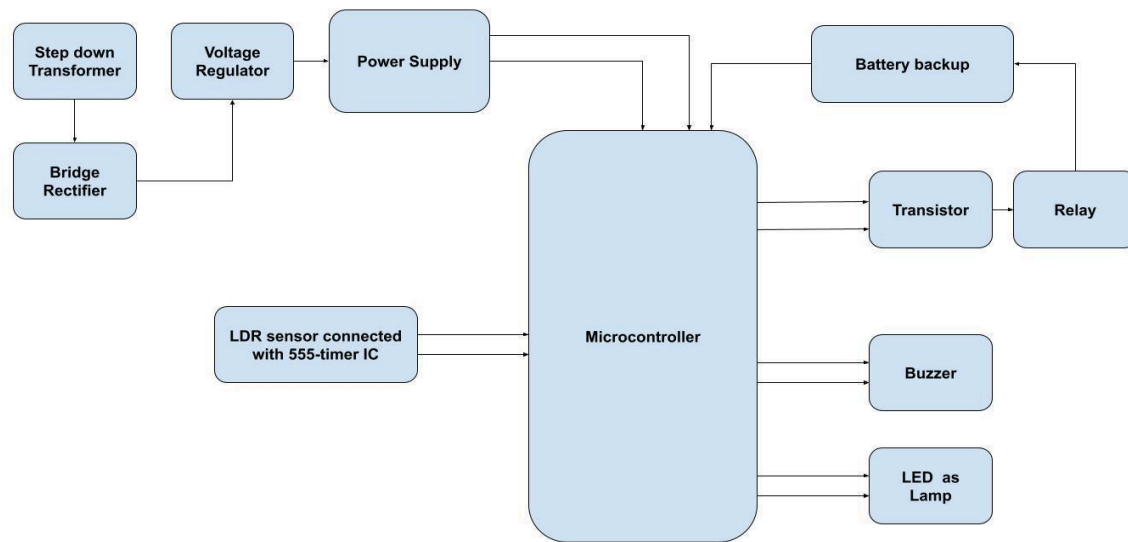


Fig. 1: Functional block diagram

As per the above diagram I described the system operation in which when the daylight would be detected by the sensor in the morning, an alarm integrated into the system would be triggered. For the control of the system, an ATMEGA328P controller IC acts as the microcontroller which I implemented with proper pin-configuration and programming code that I developed in Arduino IDE using embedded C language. Using an electronic circuit design tool I completed the schematic design of the system and then simulation conducted for the evaluation of circuit operations. After this project prototype developed with assembly of hardware project components on breadboard.

CE 3.6

Using an electronic circuit design tool I completed the schematic design of the system by integrating different components such as LDR sensor, Relay, 555-timer IC, buzzer, resistors, LEDs, capacitors, transistors, diodes, however I added the library file for ATMEGA328P controller IC and configuration carried out as per its technical specification. Also I connected the LED under the project as the night lamp. Technically I referred to the operational characteristics of the LDR as it senses the light intensity corresponding to which the resistance changes and this variation in resistance i.e. the LDRs resistance increases during darkness and it decreases on detecting full illumination. So, I deployed the LDRs for sensing the light intensity and through a

timer (555-IC) i.e. configured in monostable mode I connected the LDR with the controller IC. In order to carry out the programming of the microcontroller firstly I prepared an algorithm based on the system operations that I shown in figure 2 below as-

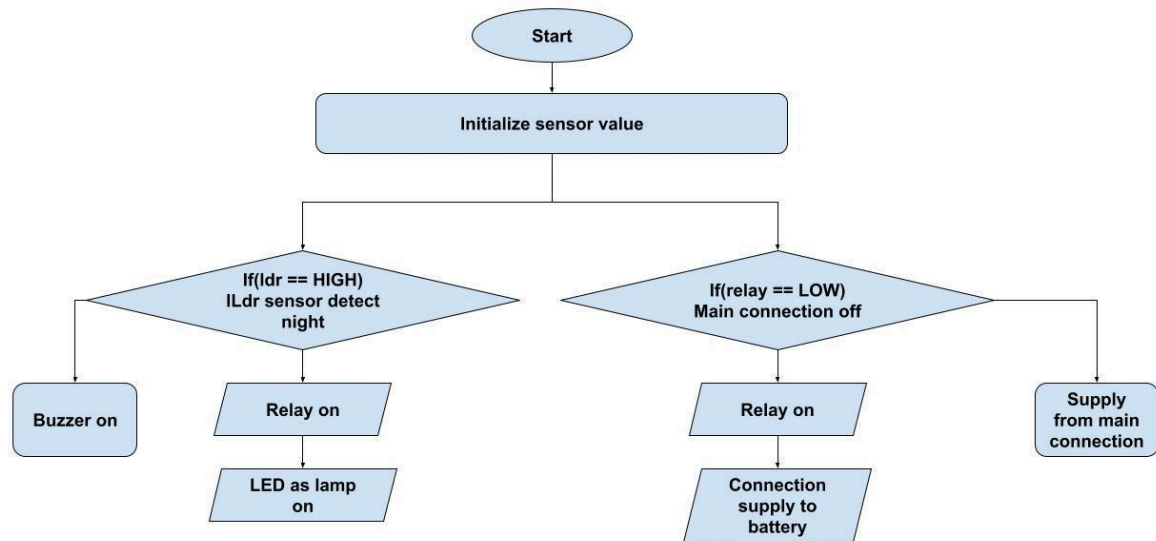


Fig. 2: Operational flowchart

As per the flowchart I program the ATMEGA328P controller IC under Arduino IDE by applying the concepts of embedded C language and all essential header files included as per the system requirements, however programming environment part with compilation process depicted in below figures-

```

int buzz_fun = 2;
int night_lmp = 3;
int lmp_led = 13;
int lmp_val;
void setup()
{
    Serial.begin(9600);
    pinMode(buzz_fun, OUTPUT);
    pinMode(night_lmp, INPUT);
    Serial.print("Automated night lamp");
    digitalWrite(buzz_fun, LOW);
    digitalWrite(lmp_led, HIGH);
}

void loop()
{
    lmp_val = digitalRead(night_lmp);
    Serial.print(lmp_val);
    delay(100);
    if (lmp_val == HIGH)
    {
        // Turn on the buzz and the LED
    }
}

```

Done compiling.

Fig. 3: Programming environment

```

void loop()
{
    lmp_val = digitalRead(night_lmp);
    Serial.print(lmp_val);
    delay(100);
    if (lmp_val == HIGH)
    {
        digitalWrite(buzz_fun, HIGH);
        digitalWrite(lmp_led, LOW);
        Serial.print("Ldr detect night");
    }

    if (lmp_val == LOW)
    {
        digitalWrite(buzz_fun, LOW);
        digitalWrite(lmp_led, HIGH);
        Serial.print("Ldr detect day");
    }
}

```

Done compiling.

Fig. 4: Programming environment

With this approach I implemented the microcontroller in terms of its pin-configuration under proteus and finally the schematic design part was completed. Then simulation I carried out in Proteus simulator that I depicted as follows-

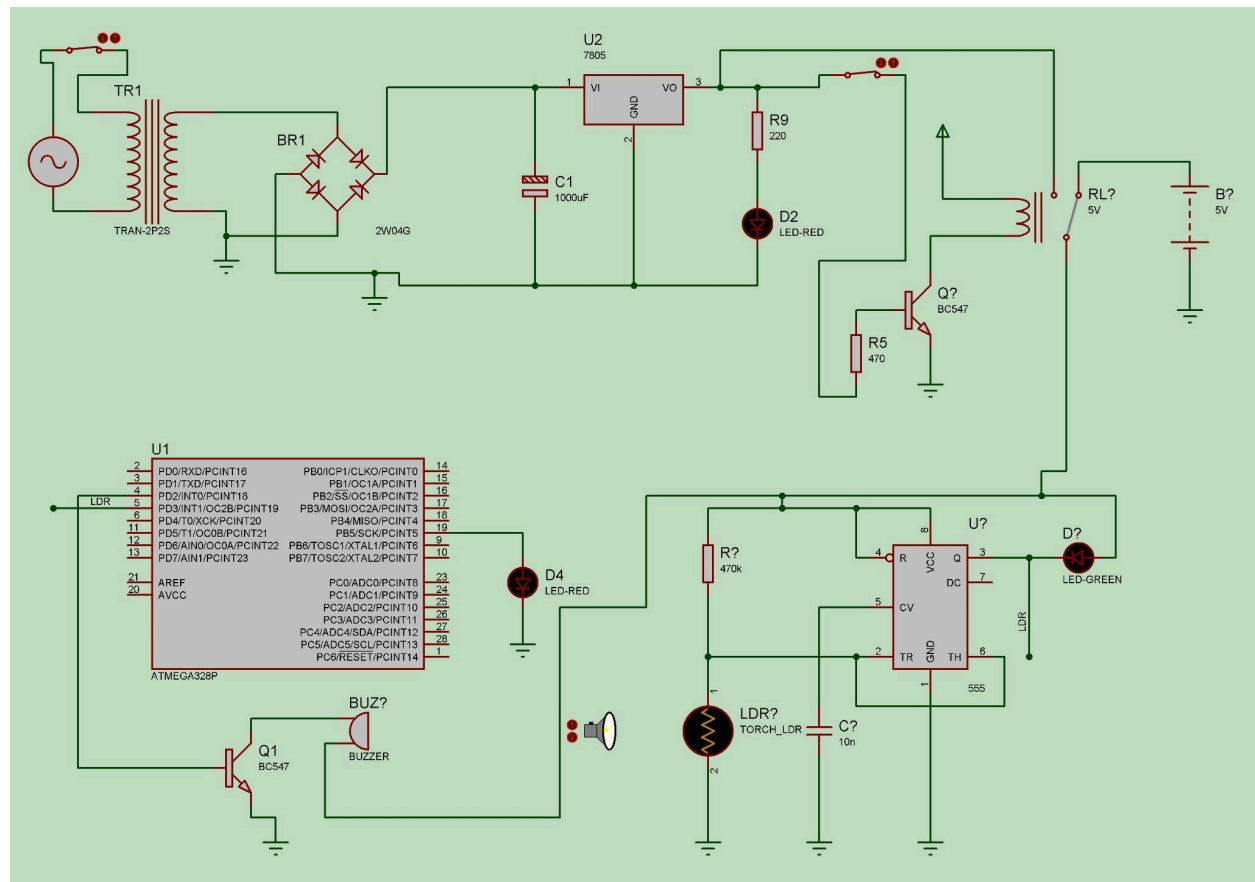


Fig. 5: Circuit configuration and simulation

For conducting the simulation I provided power to the system and for this I designed a power supply circuit by integrating the transformer (step-down type), bridge-type full wave rectifier using diodes for rectification, and the regulator IC for providing the regulated DC to the controller circuit as depicted in figure 2. Additionally in the circuit, I also deployed a battery for backup and continuous supply to a power circuit in case of failure of mains. On the availability of the main supply, the battery would get charged through the provided diode and resistor based on requirements. In case of mains failure, the battery would take the load automatically without delay whereas I provided a diode to prevent the discharge of the battery. Therefore with this circuit configuration I performed the simulation and connected Red color LED turned on to indicate the power supply and respectively system initialized as illustrated in figure 3-

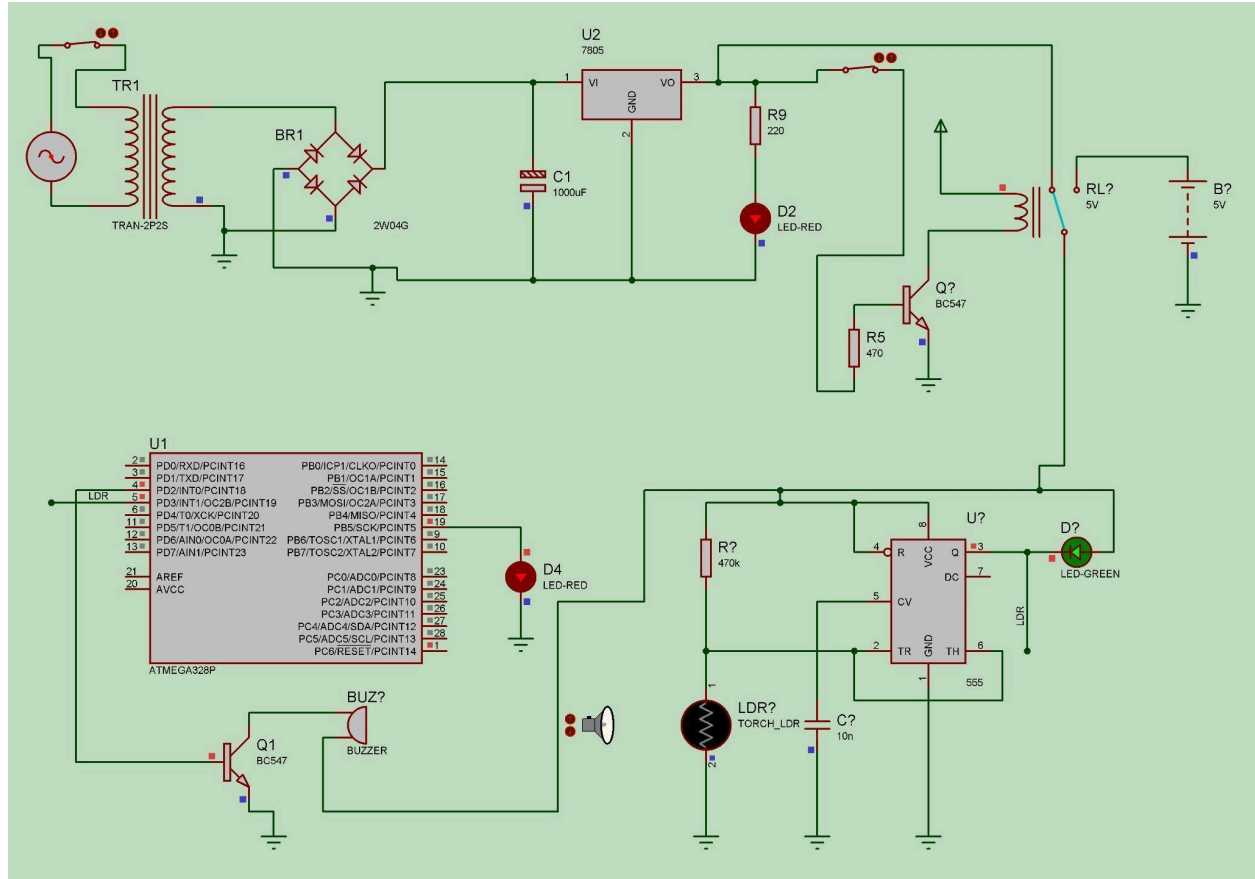


Fig. 6: Circuit configuration and simulation

After the initialization of the system the initial LDR illuminated with the room's ambient light, its resistance decreases and this would trigger the timer IC at positive potential. Moreover it results in low output at the IC and thus LED remains in off state. On the other hand the device resistance increased with a decrease in the illumination of the initial LDR. I deployed a low-value capacitor for maintaining a monostable mode of timer IC for the constant operations, due to this circuit configuration I eliminated the timer effect of the IC. Specifically I connected the buzzer with the controller IC through a transistor and further it acts as an alarm in the designed system and furthermore the related circuit simulation figures presented as follows-

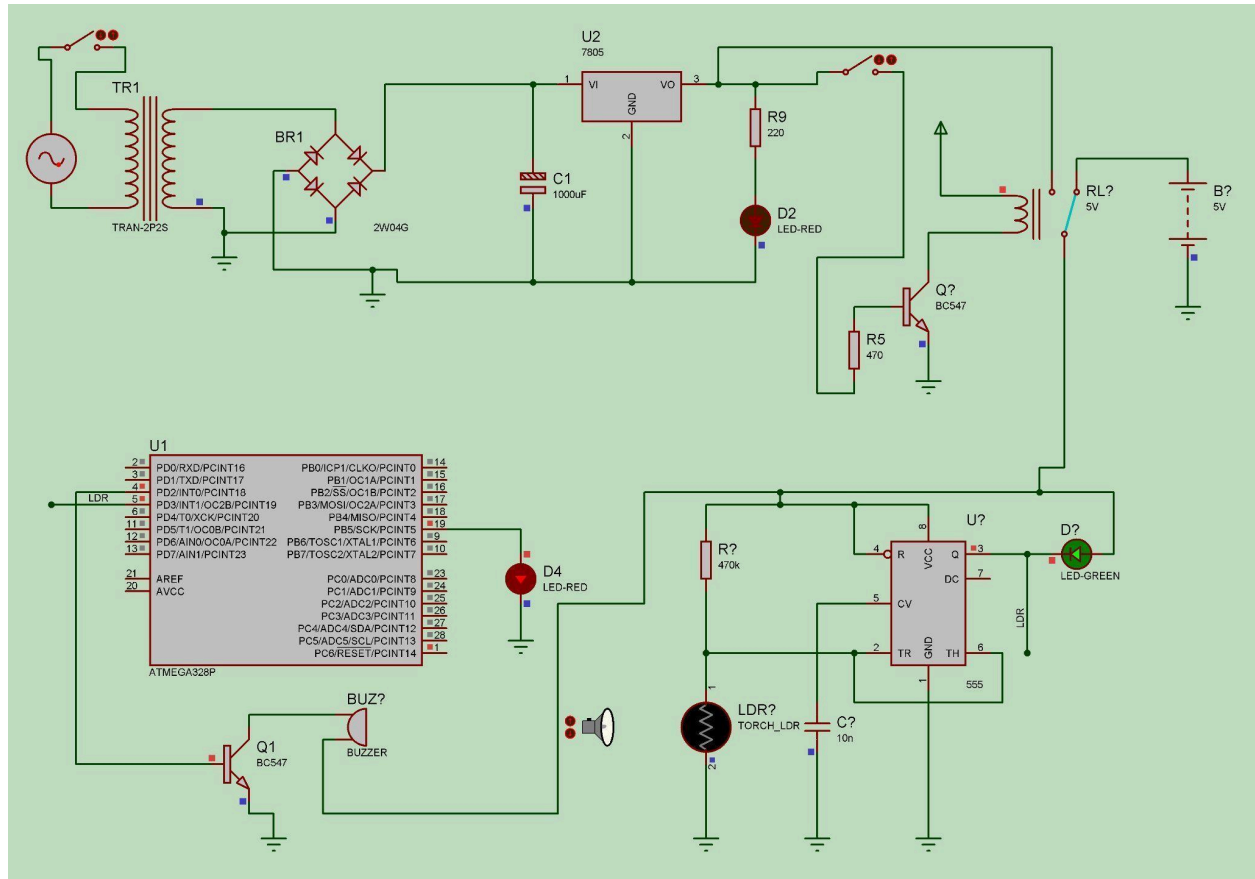


Fig. 7: Circuit configuration and simulation

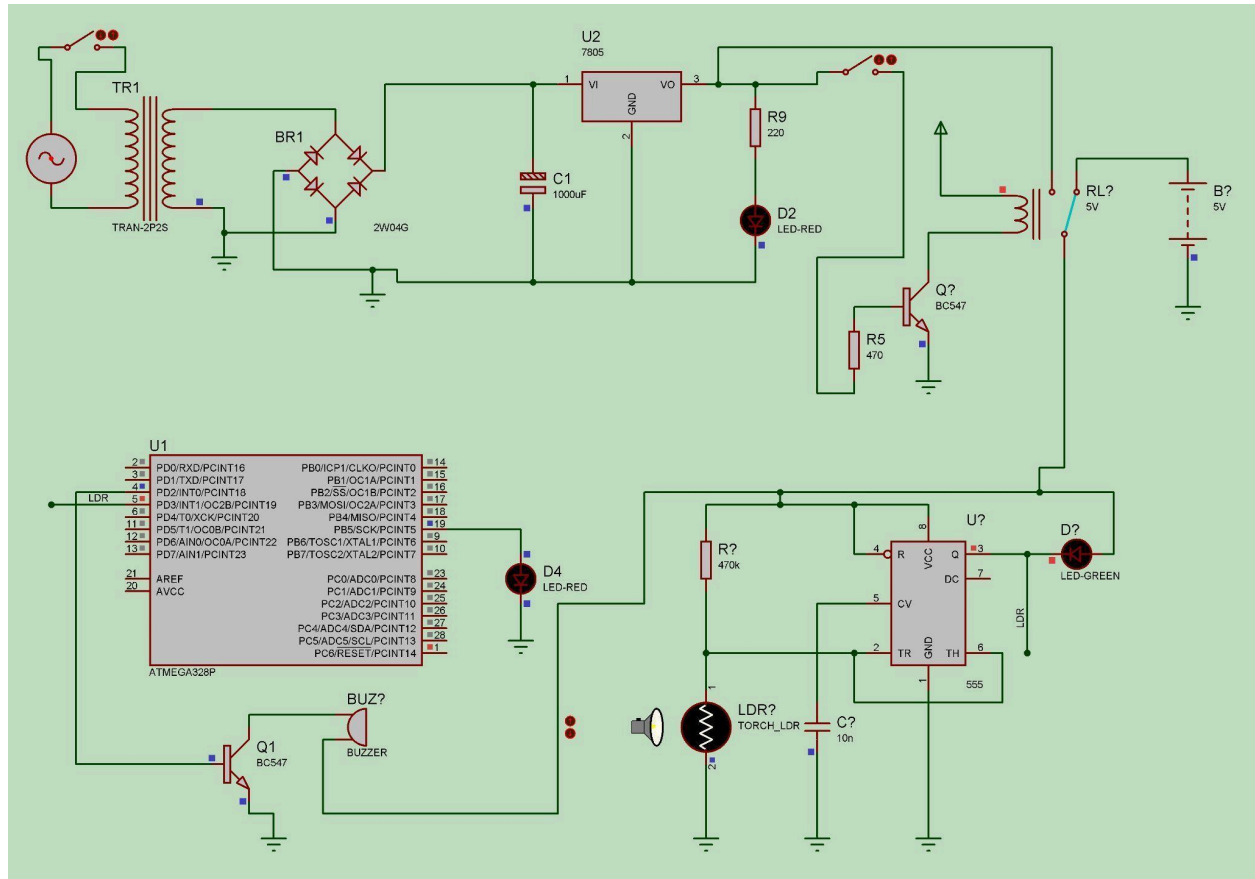


Fig. 8: Circuit configuration and simulation

Overall I designed the system wherein the programming of the microcontroller (ATMEGA328P) IC carried out to configure its pin values to connect the buzzer, LED lamp, and the light sensor with the microcontroller and achieved the optimized system design after working meticulously and using the proper design approach.

CE 3.7

Through the simulation process I analyzed the circuit performance using the Proteus simulator and for this I applied the power needed to the designed circuit corresponding to which the connected circuit components operate whereas to illustrate the working of the developed prototype model all components assembled on breadboard for temporary circuit connections and several jumper wires used to accomplish it. After providing the power I initialized the prototype model that detects light intensity for day or night and on detecting the day, the system would switch off the night lamp and further ON the alarm, however on detection of night, the system would turn ON the lights of the room. This working operations I illustrated here as follows-

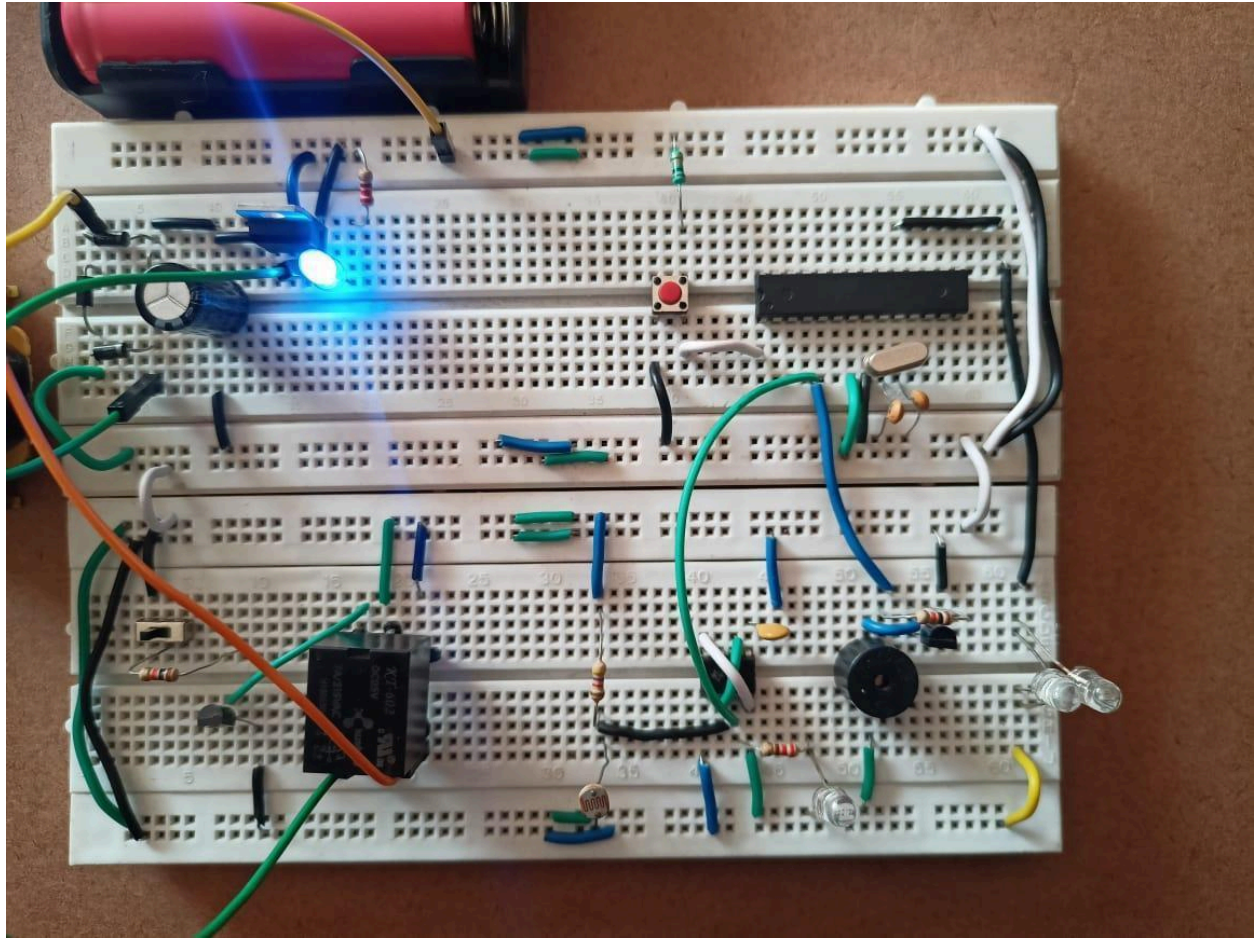


Fig. 9: Interfacing of circuit components circuit assembly and demonstration

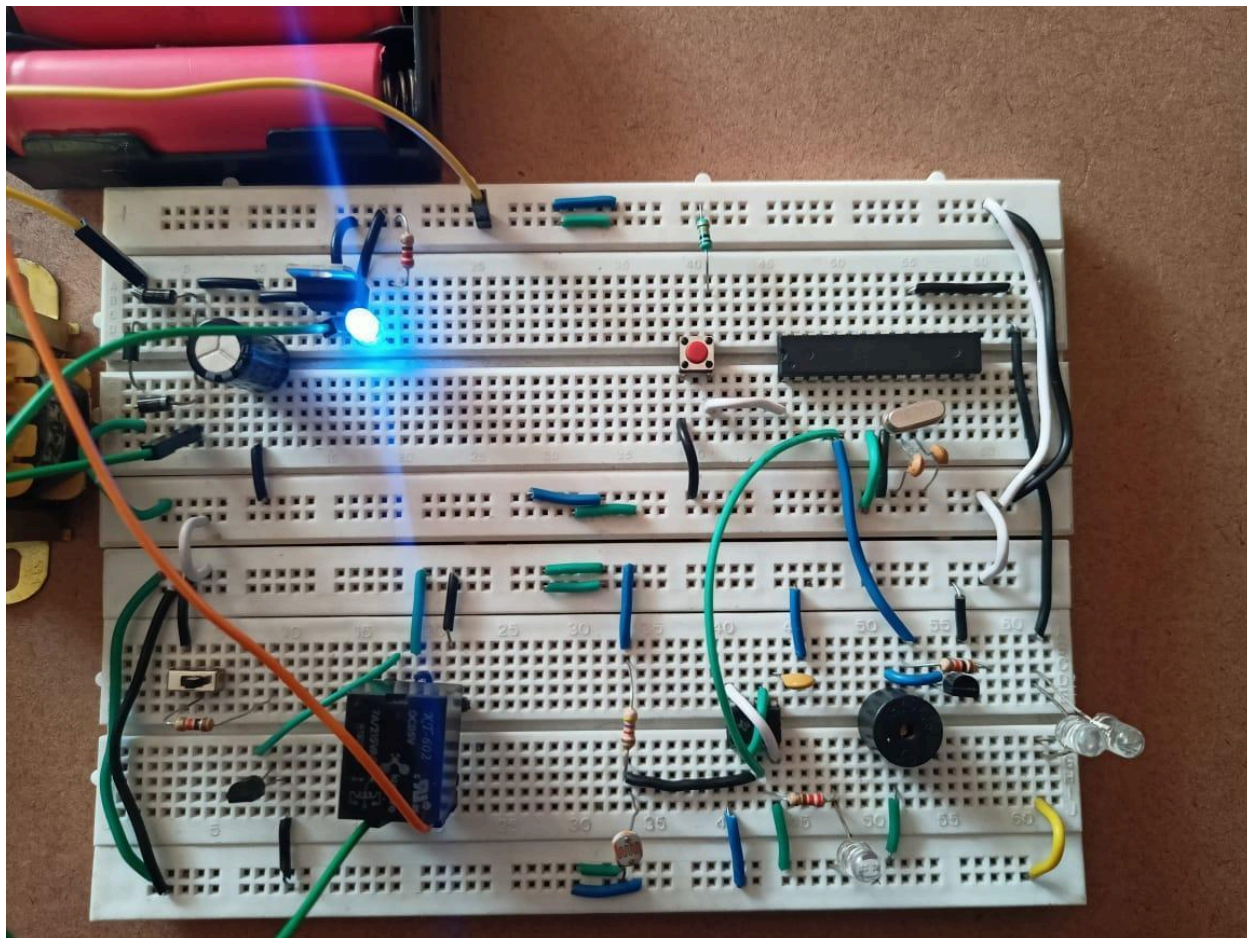


Fig. 10: Hardware circuit assembly and demonstration

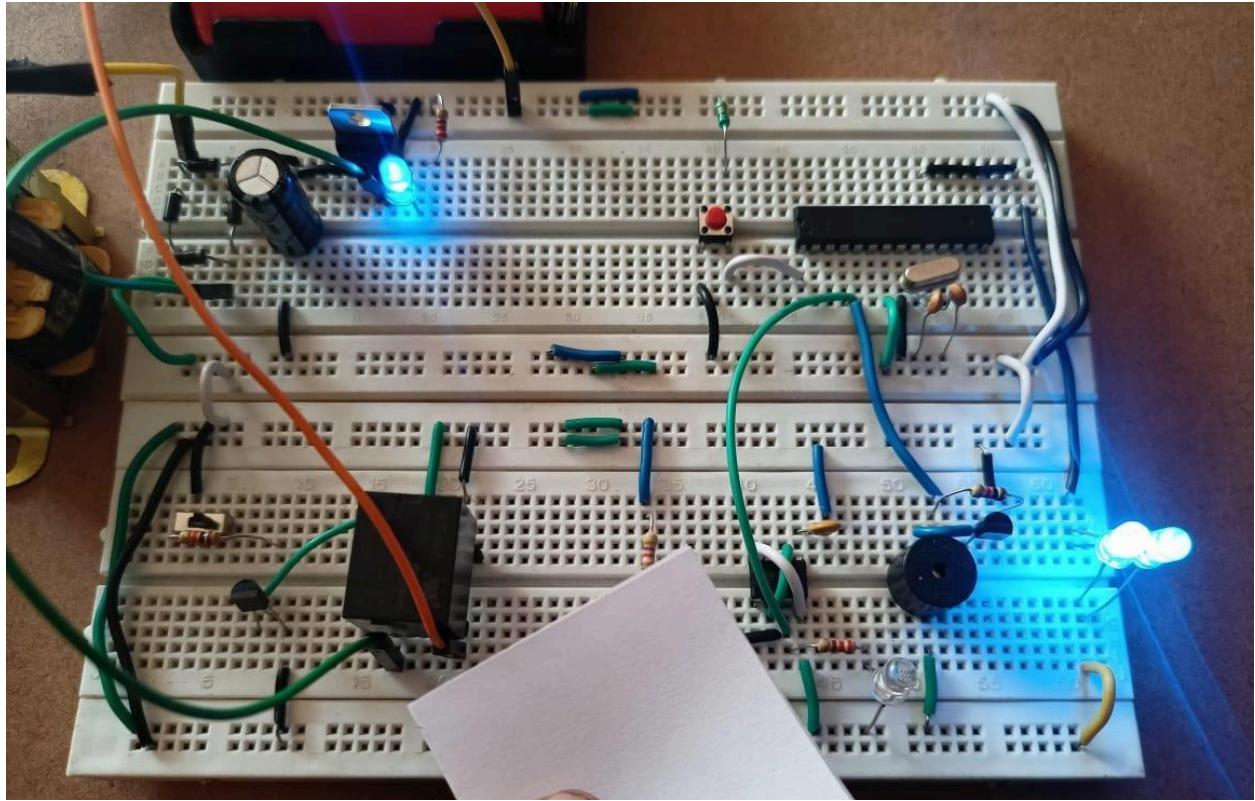


Fig. 11: Interfacing of circuit components circuit assembly and demonstration

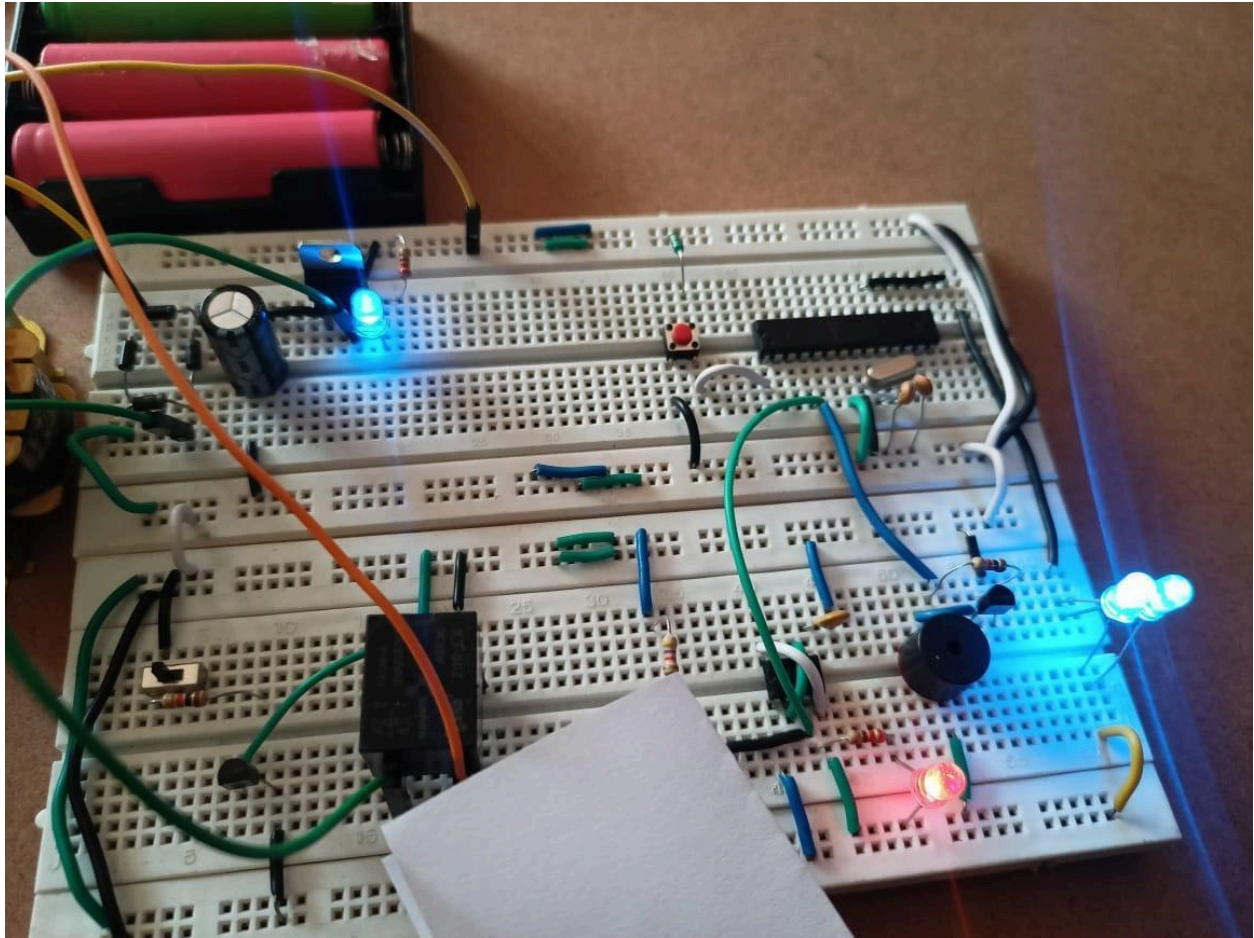


Fig. 12: Hardware circuit assembly and demonstration

For the purpose of demonstrating the hardware circuit assembly I considered a paper by which I covered the LDR sensor as depicted in figure 11 & 12 and with the variations in the light intensity I observed the switching on-off characteristics of the connected LED which acts as the lamp. In this manner I finally accomplished the hardware circuit assembly and demonstration part.

CE 3.8

I effectively developed a prototype of the LDR and microcontroller configured automated type night lamp for the room however standard design approaches and engineering fundamentals applied as per the defined plan. Under the designing process I found difficulty in designing a light sensing circuit so I referred to the timer IC (555 IC) and configured it in monostable mode to connect with a LDR along with other circuit elements in such a way that the variations in the resistance value of LDR converted into the voltage i.e. given to the IC. Further the LDR sensor

along with other electronic components integrated into the system were properly connected to achieve an optimal system. I presented the detailed system framework with accuracy and no errors were ensured. To handle the technicalities faced while the work, I deployed the best possible approaches and technologies.

D. SUMMARY

CE 3.9

I made use of my knowledge and skills to attain excellent results and to complete the project work efficiently. I applied my conceptual knowledge along with theoretical knowledge to commence the project easily and to understand the fundamentals pertaining to the engineering project. I formed action plans by utilizing my strategic thinking and systematically conducted the work. I took part in meetings to increment my technical skills and knowledge for dealing with the technicalities encountered during the work. The latest methods were used by me to carry out the tasks smoothly in accordance with the project scope. I researched different alternatives to tackle the confronted obstacles with the help of my engineering knowledge. I proficiently utilized various procedures during the work with my practical knowledge and skills. The project's aim was achieved as per the requirements and I discharged my accountabilities very well.

CE 3.10

Utilizing my theoretical knowledge and electronics engineering skills, I worked under present work and presented the system design whose primary component was microcontroller (ATMEGA328P) as overall night lamp control was based on the controller. The automated system was designed in a manner that wastage of electricity could be prevented optimally. I developed a prototype of the LDR and microcontroller configured automated type night lamp system for the room and accordingly performed necessary actions. I detailed the components deployed in project design along with their framework and reasoned out their application. I gained significant knowledge concerning the project and my engineering field which would assist in conducting similar work and meeting my study-related requirements proficiently. My

writing ability helped me present the work through its optimized documentation. I submitted the work to the department and fulfilled the study requirements.